

How to use these token cutouts

These pages contain the text *Green Eggs and Ham* by Dr Seuss. The **red text** above each token shows its prefix—the 1 word before it in the original text.

Training: build the cutouts model

1. **cut out the tokens** from the sheets above along the dotted lines (carefully—keep each prefix-token pair together)
2. **spread them out** on a table, face up, no overlap if possible
3. that's your trained language model—every prefix-token pair from your training text is now a physical cutout on the table

Optional extension: once you've got the hang of it, group the cutouts into piles by their **red prefix label**. Each pile is a “bucket” in computer-science speak—the same trick a hash table uses to find values quickly.

Generation: sample new text

1. **choose a starting prefix**—any word that appears as a **red label** on at least one cutout
2. **write it down**, then make it your “current word”
3. **find candidates**—look across the spread for cutouts whose red prefix label matches your current word
4. **pick one** visually (your eye will land on cutouts more often when there are more of them—that's weighted sampling for free)
5. **write down** the picked cutout's token, then **put it back** in the spread
6. the token you just wrote is your new current word—go back to step 3

Tip: if a prefix has more cutouts in the spread, words following it are more common in the source text, and your eye will pick them more often.

Model statistics

- **Total tokens:** 940 (940 kept)
- **Entropy:** 1.75 bits/token — how unpredictable each pick is
- **Perplexity:** 3.4 — effective number of choices per pick

I am Sam I am Sam

Sam I am that Sam I am

that Sam I am I do not like

that Sam I am would you

you like green eggs and ham I do

do not like them Sam I am

I do not like green eggs and

and ham would you like them

them here or there I would not

not like them here or there I

I would not like them anywhere

anywhere I do not like green eggs

eggs and ham I do not like

like them Sam I am would

would you like them in a house

house would you like them with a

a mouse I do not like them in

in a house I do not like them

them with a mouse I do not

not like them here or there I

I do not like them anywhere

anywhere I do not like green eggs

eggs and ham I do not like

like them Sam I am would

would	you	eat	them	in	a	box	would
165	166	167	168	169	170	171	171
would	you	eat	them	with	a	fox	not
172	173	174	175	176	177	178	178
not	in	a	box	not	with	a	fox
179	180	181	182	183	184	185	187
not	in	a	house	not	with	a	with
188	189	190	191	192	193	194	195
a	mouse	I	would	not	eat		
196	197	198	199	200	201		
eat	them	here	or	there	I	would	
202	203	204	205	206	207	208	208
would	not	eat	them	anywhere	I		
209	210	211	212	213	214		
I	would	not	eat	green	eggs	and	
215	216	217	218	219	220	220	
and	ham	I	do	not	like	them	
221	222	223	224	225	226	227	228
Sam	I	am	would	you	could		
229	230	231	232	233	234	235	235
could	you	in	a	car	eat	them	
236	237	238	239	240	241	242	
eat	them	here	they	are	I	would	
243	244	245	246	247	248	249	249

would not , could not , in a car
not 250 251 252 253 254 255 256 257 258

, you may like them you
259 260 261 262 263 264

you will see you may like them
265 266 267 268 269 270 271

them in a tree I would not
272 273 274 275 276 277 278

, could not in a tree not in
279 280 281 282 283 284 285 286

in a car you let me be I do
287 288 289 290 291 292 293 294 295

do not like them in a box I
296 297 298 299 300 301 302 303

I do not like them with a fox
304 305 306 307 308 309 310

fox I do not like them in a
311 312 313 314 315 316 317 318

a house I do not like them
319 320 321 322 323 324 325

them with a mouse I do not
326 327 328 329 330 331 332

not like them here or there I
333 334 335 336 337 338 339

I do not like them anywhere

340

341

342

343

344

anywhere



I

do

not

like

green

I do not like green eggs

345

346

347

348

349

350

351

eggs

and

ham



I

do

not

and ham I do not like

352

353

354

355

356

357

358

like

them



Sam

I

am



a

them Sam I am a train

359

360

361

362

363

364

365

366

train

a

train

a

train

a

train

a train a train a train could

367

368

369

370

371

372

373

could

you



would

you

on

a

you would you on a train

374

375

376

377

378

379

380

train

not

on

a

train

not

in

a

not on a train not in a tree

381

382

383

384

385

386

387

388

tree

not

in

a

car

Sam

let

me

be

not in a car Sam let me be I

389

390

391

392

393

394

395

396

397

I

would

not



could

not



in

would not could not in a

398

399

400

401

402

403

404

405

a

box



I

would

not



box I would not could

406

407

408

409

410

411

412

could

not



with

a

fox



I

not with a fox I will

413

414

415

416

417

418

419

420

will

not

eat

them

in

a

house



not eat them in a house I

421

422

423

424

425

426

427

428

I	will	not	eat	them	with	
will	not	eat	them	with	a	
429	430	431	432	433	434	
a	mouse	□	I	will	not	eat
mouse	□	I	will	not	eat	them
435	436	437	438	439	440	441
them	here	or	there	□	I	will
here	or	there	□	I	will	not
442	443	444	445	446	447	448
eat	them		anywhere	□	I	do
them	anywhere	□	I	do	not	
450	451	452	453	454	455	
not	eat	green	eggs	and	ham	□
eat	green	eggs	and	ham	□	I
456	457	458	459	460	461	462
I	do	not	like	them	□	Sam
do	not	like	them	□	Sam	I
463	464	465	466	467	468	469
I	am	□	say	in	the	dark
am	□	say	in	the	dark	here
470	471	472	473	474	475	476
in	the	dark	would	you	□	
the	dark	would	you	□	could	
478	479	480	481	482	483	
could	you	□	in	the	dark	I
you	□	in	the	dark	I	would
484	485	486	487	488	489	490
would	not	□	could	not	□	in
not	□	could	not	□	in	the
491	492	493	494	495	496	497
dark	□	would	you	□	could	you
□	would	you	□	could	you	□
499	500	501	502	503	504	505
□	in	the	rain	I	would	not
in	the	rain	I	would	not	□
506	507	508	509	510	511	512

,	could	not	in	the	rain	.	not
513	514	515	516	517	518	519	
not	in	the	dark	.	not	on	a
520	521	522	523	524	525	526	527
train	.	not	in	a	car	.	not
528	529	530	531	532	533	534	535
	not	in	a	car	.	not	in
528	529	530	531	532	533	534	535
a	tree	.	I	do	not	like	them
537	538	539	540	541	542	543	544
tree	.	I	do	not	like	them	.
537	538	539	540	541	542	543	544
.	Sam	.	you	see	.	not	in
545	546	547	548	549	550	551	552
Sam	.	you	see	.	not	in	a
545	546	547	548	549	550	551	552
a	house	.	not	in	a	box	.
553	554	555	556	557	558	559	560
house	.	not	in	a	box	.	not
553	554	555	556	557	558	559	560
not	with	a	mouse	.	not	with	a
561	562	563	564	565	566	567	
with	a	mouse	.	not	with	a	
561	562	563	564	565	566	567	
a	fox	.	I	will	not	eat	
568	569	570	571	572	573	574	
fox	.	I	will	not	eat		
568	569	570	571	572	573	574	
them	here	or	there	.	I	do	not
575	576	577	578	579	580	581	582
here	or	there	.	I	do	not	like
575	576	577	578	579	580	581	582
like	them	anywhere	you	do	not		
583	584	585	586	587			
like	green	eggs	and	ham	I	do	
588	589	590	591	592	593	594	
like	green	eggs	and	ham	I	do	
588	589	590	591	592	593	594	
do	not	like	them	.	Sam	I	am
595	596	597	598	599	600	601	602
not	like	them	.	Sam	I	am	.
595	596	597	598	599	600	601	602

could you could you would you

with a goat I would not

could not with a goat would

would you could you on a boat

I could not would not

on a boat I will not will

will not with a goat I will

will not eat them in the rain

not in the dark not in a tree

tree not in a car you let me be I

I do not like them in a box

I do not like them with a

a fox 693	fox 694	I 695	will 696	not 697	eat 698	them 699	in 700
in 701	a 702	house 703	I 704	do 705	not 706	like 707	them 708
them 709	with 710	a 711	mouse 712	I 713	do 714	not 715	
not 716	like 717	them 718	here 719	or 720	there 721	I 722	
I 723	do 724	not 725	like 726	them 727			
anywhere 728	I 729	do 730	not 731	like 732	green 733	eggs 734	
eggs 735	and 736	ham 737	I 738	do 739	not 740	like 741	
them 742	Sam 743	I 744	am 745	you 746	do 747	not 748	
not 750	like 751	them 752	so 753	you 754	say 755	try 756	
try 758	them 759	try 760	them 761	and 762	you 763		
may 764	try 765	them 766	and 767	you 768	may 769	I 770	
I 772	say 773	Sam 774	If 775	you 776	let 777	me 778	be 779

be 780	• 781	I 782	will 783	try 784	them 785	• 786	you 787
I	will	try	them	•	you	will	
will 788	see 789	• 790	• 791	• 792	and 793	he 794	tries 795
see	•	•	•	and	he	tries	
tries 796	them 797	• 798	• 799	say 800	I 801	like 802	green 803
them	•	•	•	say	I	like	green
green 804	eggs 805	and 806	ham 807	I 808	do 809	I 810	like 811
eggs	and	ham	I	do	I	like	
like 811	them 812	• 813	Sam 814	I 815	am 816	and 817	
them	•	Sam	I	am	and	I	
I 818	would 819	eat 820	them 821	in 822	a 823	boat 824	• 825
would	eat	them	in	a	boat	•	
• 825	and 826	I 827	would 828	eat 829	them 830	with 831	a 832
and	I	would	eat	them	with	a	
a 832	goat 833	• 834	and 835	I 836	will 837	eat 838	
goat	•	and	I	will	eat		
them 839	in 840	the 841	rain 842	• 843	and 844	in 845	the 846
in	the	rain	•	and	in	the	
dark 847	• 848	and 849	on 850	a 851	train 852	• 853	and 854
•	and	on	a	train	•	and	in
a 856	car 857	• 858	and 859	in 860	a 861	tree 862	• 863
car	•	and	in	a	tree	•	
they 864	are 865	so 866	good 867	• 868	so 869	good 870	• 871
are	so	good	•	so	good	•	

, you see so I will eat them in
871 872 873 874 875 876 877 878

in a box and I will eat them
879 880 881 882 883 884 885 886

them with a fox and I will eat
887 888 889 890 891 892 893 894

eat them in a house and I
895 896 897 898 899 900 901

I will eat them with a mouse
902 903 904 905 906 907

mouse and I will eat them here
908 909 910 911 912 913 914

here and there say I will eat
915 916 917 918 919 920 921

eat them anywhere I do so like
922 923 924 925 926 927

like green eggs and ham Thank
928 929 930 931 932

Thank you Thank you Sam I am
933 934 935 936 937 938 939

am
940